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https://drive.google.com/file/d/1I3l_qEAKA8BTP8DP00sZ-tdZ5K8ct-JQ/view?usp=sharing [slide 40-42]

Notation:

m = Number of training examples

x 's = "input" variable/features

y 's = output variable/target variable

(x, y) : a training example

(x^i, y^i) : the i^{th} training example

★ linear regression:

Linear Regression is a method to model the relationship between:

- Input (x) → **predictor** variable
- Output (y) → **response** variable

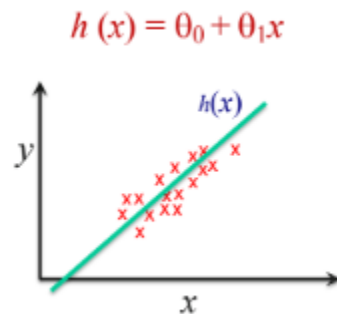
👉 It fits a **straight line** to the data



Basic Equation



- **w_0 (intercept)**: value of y when $x = 0$
- **w_1 (slope)**: how much y changes when x increases



Another formula:

$$h(x) = \theta_0 + \theta_1 x$$

- **$h(x)$ = predicted value**

Example

- x = house size
- y = price

 Model:

- $y = 50 + 2x$

Meaning:

- Base price = 50
- Each unit increase in size $\rightarrow$ price increases by 2

Method of Least Squares

Choose line that **minimizes error** between:

- **Actual** values (y)
- **Predicted** values ($\hat{y}$)



Error

- Error = (Actual - Predicted)



Goal

Minimize total squared error:

$$\sum (y_i - \hat{y}_i)^2$$

👉 Squaring:

- **Avoids negative cancellation**
- **Penalizes large errors more**



Multiple/Multivariate Linear Regression

- **multiple input features**
- Instead of 1 variable (x), we use many: $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n$

Suppose we have the following features

x_1	x_2	x_3	x_4	y
Size ft ²	bedrooms	floors	Age	Price
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
...	...	...	...	...

A hypothesis function $h(x)$ might be:

$$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

or $h(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n \quad (x_0=1)$

e.g., $h(x) = 80 + 0.1 \cdot x_1 + 0.01 \cdot x_2 + 3 \cdot x_3 - 2 \cdot x_4$

👉 So price depends on ALL of these features



Equation

$$y = w_0 + w_1 x_1 + w_2 x_2$$

$$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

- $h(x) \rightarrow$ predicted value (price)
- $\theta_0 \rightarrow$ intercept (base value)
- $\theta_1, \theta_2, \dots \rightarrow$ weights (importance of each feature)
- $x_1, x_2, \dots \rightarrow$ input features

👉 More features $\rightarrow$ better prediction (if used properly)

$$h(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

👉 Where:

- $x_0 = 1$ (just for mathematical convenience)
-

Training Data Format

- Data looks like:
 - $(x_1, y_1), (x_2, y_2), \dots$

👉 For multiple inputs:

- $(X_1, y_1), (X_2, y_2), \dots$

Where:

- **X** = **vector** of **features** $(x_1, x_2, \dots)$
-

How It Is Solved

- **Extend** least squares method
 - Or use tools like:
 - SAS
 - S-Plus
 - (modern: Python, sklearn, etc.)
-

Nonlinear Relationships

Some nonlinear functions can be **converted to linear form**

Polynomial Linear Regression

Idea

- Still **linear regression**
- But uses **polynomial features** ($x^2, x^3, \dots$)

👉 Helps model **curved (nonlinear)** relationships

Hypothesis Function (Polynomial)

$$h(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2^2 + \theta_3 x_3^3 + \dots + \theta_n x_n^n$$

The Cost Function

We want to choose parameters (θ_0, θ_1) such that:

👉 Our **prediction** $h(x)$ is **as close as possible** to **actual** value y

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h(x^{(i)}) - y^{(i)})^2$$

- $J(\theta_0, \theta_1) \rightarrow$ cost (error)
 - $m \rightarrow$ number of training examples
 - $h(x^{(i)}) \rightarrow$ predicted value
 - $y^{(i)} \rightarrow$ actual value
-

Formulas for Linear Regression

$$h(x) = \theta_0 + \theta_1 x$$

$$\theta_1 = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$\theta_0 = \frac{\sum y - \theta_1 \sum x}{n}$$

- $\theta_1 \rightarrow$ controls slope
- $\theta_0 \rightarrow$ shifts line up/down

› ass 2 Q6

Question 6:

[10 Marks]

Ecology explores relationships between organisms and other living (biotic) things or non-living (abiotic) components in their environment. Certain factors may impact the abundance, distribution, or physiology of organisms, including abiotic factors such as temperature, moisture, or sunlight, and biotic factors such as the presence of predators or competitors. Linear regressions can be used to quantify these relationships and predict organism responses to various levels of the abiotic or biotic factor. Ecologists tested the grazing pressure of green crabs on clams. They constructed $1m^2$ cages and planted 300 clams in each cage. Two days later, they counted the number of remaining clams and recorded the data in the data below.

Number of Crabs	Number of Clams Remaining
2	137
4	70
2	184
5	0
4	35
0	297
3	122
5	1
1	253
3	150

First, perform a linear regression using the step-by-step instructions for calculating m (slope) and b (intercept) of the regression line. **What is the full equation for the regression line that you calculated?**

Step 1. Decide which variable is the independent variable and which is the dependent variable: In this example, the number of crabs is the independent variable (x), and the number of clams remaining is the dependent variable (y). This is because the researchers chose the number of crabs to place in each plot, and the resulting number of clams *depends* on how many crabs are present.

Step 2. Calculate x^2 for every value of x , and y^2 for every value of y . For each pair of values, calculate $x \times y$.

You should end up with ten values (the same number of values of x) for x^2 , ten values for y^2 , and ten values for $x \times y$.

x^2	y^2	$x \times y$
4	18,769	274
16	4,900	280
4	33,856	368
25	0	0
16	1,225	140
0	88,209	0

9	14,884	366
25	1	5
1	64,009	253
9	22,500	450

Step 3. Calculate $\sum x$, $\sum y$, $\sum(x)^2$, $\sum(y)^2$, and $\sum(x \times y)$.

$$\sum x = 2 + 4 + 2 + 5 + 4 + 0 + 3 + 5 + 1 + 3 = 29$$

$$\sum y = 137 + 70 + 184 + 0 + 35 + 297 + 122 + 1 + 253 + 150 = 1249$$

$$\sum(x^2) = 4 + 16 + 4 + 25 + 16 + 0 + 9 + 25 + 1 + 9 = 109$$

$$\sum(y^2) = 18769 + 4900 + 33856 + 0 + 1225 + 88209 + 14884 + 1 + 64009 + 22500 = 248353$$

$$\sum(x \times y) = 274 + 280 + 368 + 0 + 140 + 0 + 366 + 5 + 253 + 450 = 2136$$

$\sum x$	$\sum y$	$\sum(x^2)$	$\sum(y^2)$	$\sum(x \times y)$
29	1249	109	248,353	2136

Step 4. Familiarize yourself with the equation of a line: $y = mx + b$.

Step 5. Calculate m (slope):

$$\frac{n(\Sigma(x \times y)) - (\Sigma x)(\Sigma y)}{n(\Sigma(x^2)) - (\Sigma x)^2}$$

To simplify this calculation, let's start with the **numerator**: $(n(\Sigma(x \times y)) - (\Sigma x)(\Sigma y))$

Remember **n**! In this example, $n = 10$ because we have 10 values of x and 10 values of y .

$$\text{numerator} = (10(2136) - (29)(1249))$$

$$\text{numerator} = (21360 - 36221)$$

$$\text{numerator} = -14861$$

Now we can calculate the bottom half of the equation (this is called the **denominator**): $(n(\Sigma(x^2)) - (\Sigma x)^2)$

$$\text{denominator} = (10(109) - (29)^2)$$

$$\text{denominator} = (1090 - 841)$$

$$\text{denominator} = 249$$

To finish the calculation, divide the numerator by the denominator:

$$m = -\frac{14861}{249} = -59.683$$

Here, we have rounded the actual answer from a long decimal string (-59.68273092...) to **-59.683**.

Step 6. Calculate b (intercept): $\frac{\Sigma y - m(\Sigma x)}{n}$

Again we can start on the **numerator** (top half of the equation): $(\Sigma y - m(\Sigma x))$

$$\text{numerator} = (1249 - (-59.683)(29))$$

$$\text{numerator} = (1249 - (-1730.807))$$

$$\text{numerator} = 2979.807$$

Next, we would calculate the **denominator** (bottom half of the equation), but this is just n !

$$\text{denominator} = 10$$

To finish the calculation, divide the numerator by the denominator:

$$b = \frac{2979.807}{10} = 297.98$$

Again, we have rounded the actual answer from a longer decimal (297.9807) to **297.98**.

Step 7. Replace ' b ' and ' m ' in the line equation: $y = mx + b$.

From the previous steps, we know:

$$m = -59.683$$

$$b = 297.98$$

So the final equation of the linear regression is: $y = -59.683x + 297.98$

Example: Linear Regression

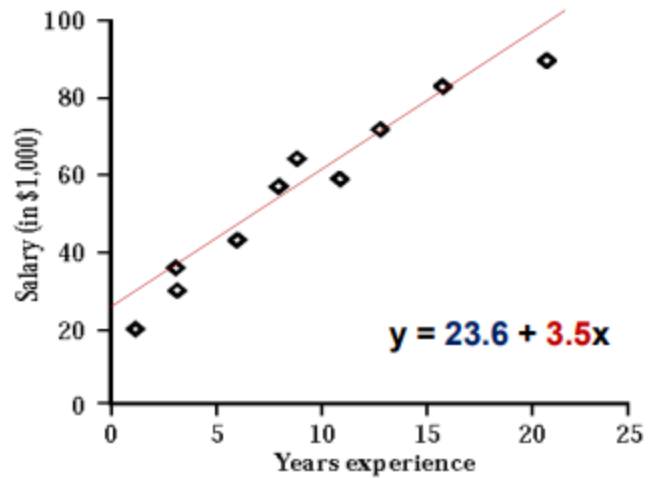
X (Years experience)	Y (Salary)
3	30K
8	57K
9	64K
13	75K
3	36K
6	43K
11	59K
21	90K
1	20K
16	83K

- Given data

$$\bar{x} = 9.1 \quad \bar{y} = 55.4$$

$$w_1 = \frac{(3-9.1)(30-55.4) + \dots + (16-9.1)(83-55.4)}{(3-9.1)^2 + \dots + (16-9.1)^2}$$

$$w_0 = 55.4 - (3.5)(9.1) = 23.6$$



Example: Linear Regression Analysis

X (Years experience)	Y (Salary)	Y (Predicted) [Y=23.6 +3.5X]	Mean Absolute Error	Root Mean Squared Error
3	30K	34.1K	abs(Y-Yp)=4.1	Sqr(Y-Yp)=16.81
8	57K	51.6K	5.4	29.16
9	64K	55.1K	8.9	79.21
13	75K	69.1K	5.9	34.81
3	36K	34.1K	1.9	3.61
6	43K	44.6K	1.6	2.56
11	59K	62.1K	3.1	9.61
21	90K	97.1K	7.1	50.41
1	20K	27.1K	7.1	50.41
16	83K	79.6K	3.4	11.56
			4.85	5.37

$$MAE = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (P_i - O_i)^2}{n}}$$

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

Where:

$\hat{y}_i$ = Predicted value for the i^{th} data point

y_i = Actual value for the i^{th} data point

n = number of observations